Exercises

CHI-SQUARE TEST & CONTINGENCY ANALYSIS

Question 1

a) A restaurant manager wants to analyze whether the distribution of customer reservations is uniform across weekdays. The manager recorded the number of reservations made each day during a randomly chosen week. The data is presented in the table below:

Day	Monday	Tuesday	Wednesday	Thursday	Friday
Reservation	30	25	35	28	40

Based on this data, conduct a hypothesis test to prove that the number of reservations is equally distributed throughout the weekdays. Use significance level, $a = 0.05$.

(Calculate with precision of 3 decimal place)

Question 2

b) A museum curator wants to investigate whether visitor attendance varies significantly across different days of the week. The curator recorded the number of visitors on each day during a randomly selected week. The data is summarized in the table below:

Day	Monday	Tuesday	Wednesday	Thursday	Friday
Visitors	500	600	550	700	800

Based on this data, conduct a hypothesis test to prove that the number of reservations is equally distributed throughout the weekdays. Use significance level, $a = 0.05$. *(Calculate with precision of 3 decimal place)*

Question 3

Internet has become the most important means of communication and gathering information. People with different age use Internet for different purposes. A random sample of 159 adults yielded the data shows in Table 1 to find out the relationship between age and the usage of internet.

Table 1

		Age	
		25- 64	65+
Internet Usage	Never	38	31
	Sometimes	31	4
	Every day	50	5

a) State the null and alternative hypotheses.
b) Calculate the expected count and the test statistic
c) At the 1% significance level, do the data provide sufficient evidence to conclude that a relationship exists between age and Internet usage?

Question 4

A case-control study was conducted to investigate a relationship between the colors of helmets worn by motorcycle drivers and whether they are injured or killed in a crash. Results are given in the Table 1.

Table 1

		Color of Helmet		
		Black	White	Red
Injuries	Controls (not injured)	491	377	170
	Cases (injured or killed)	213	112	70

a) Test the claim that injuries are independent of helmet color at 0.05 sigificance level.
b) Should motorcycle drivers choose helmets with a particular color?

THANK YOU !